ECE 314 (BTech)

A0I333 Static Gate

Group Number: 19

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Group Members

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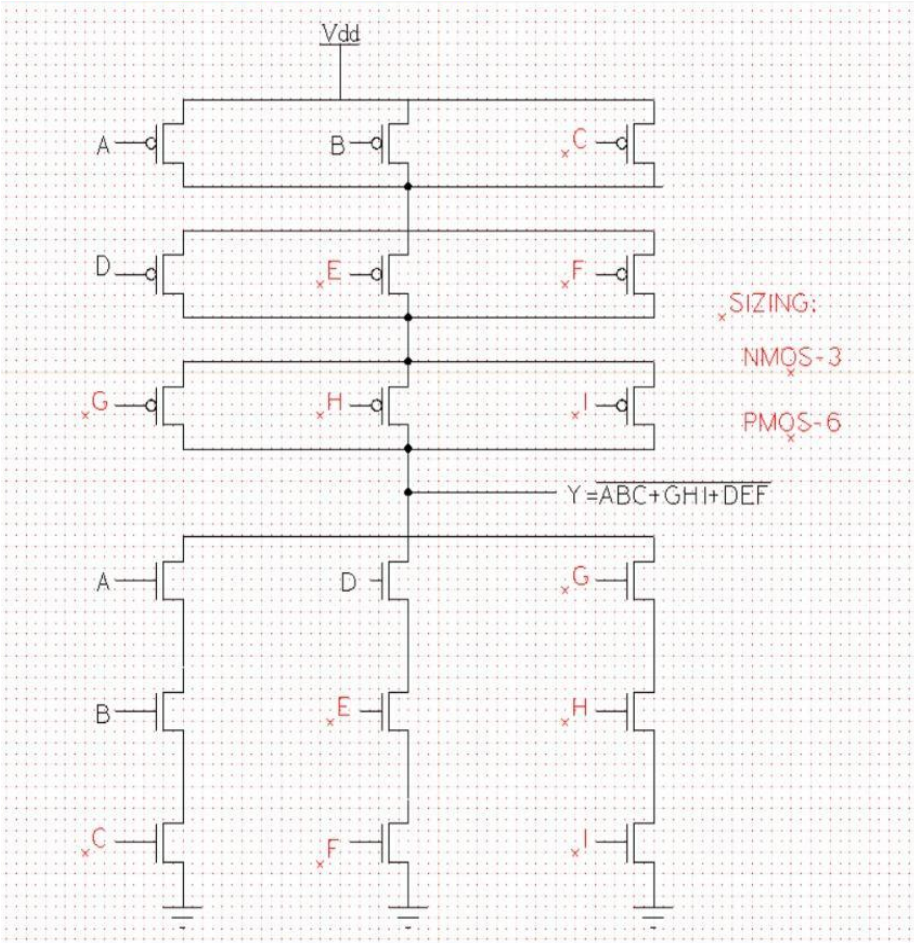
Rahul Kumar(2022391)
Sachin kumar

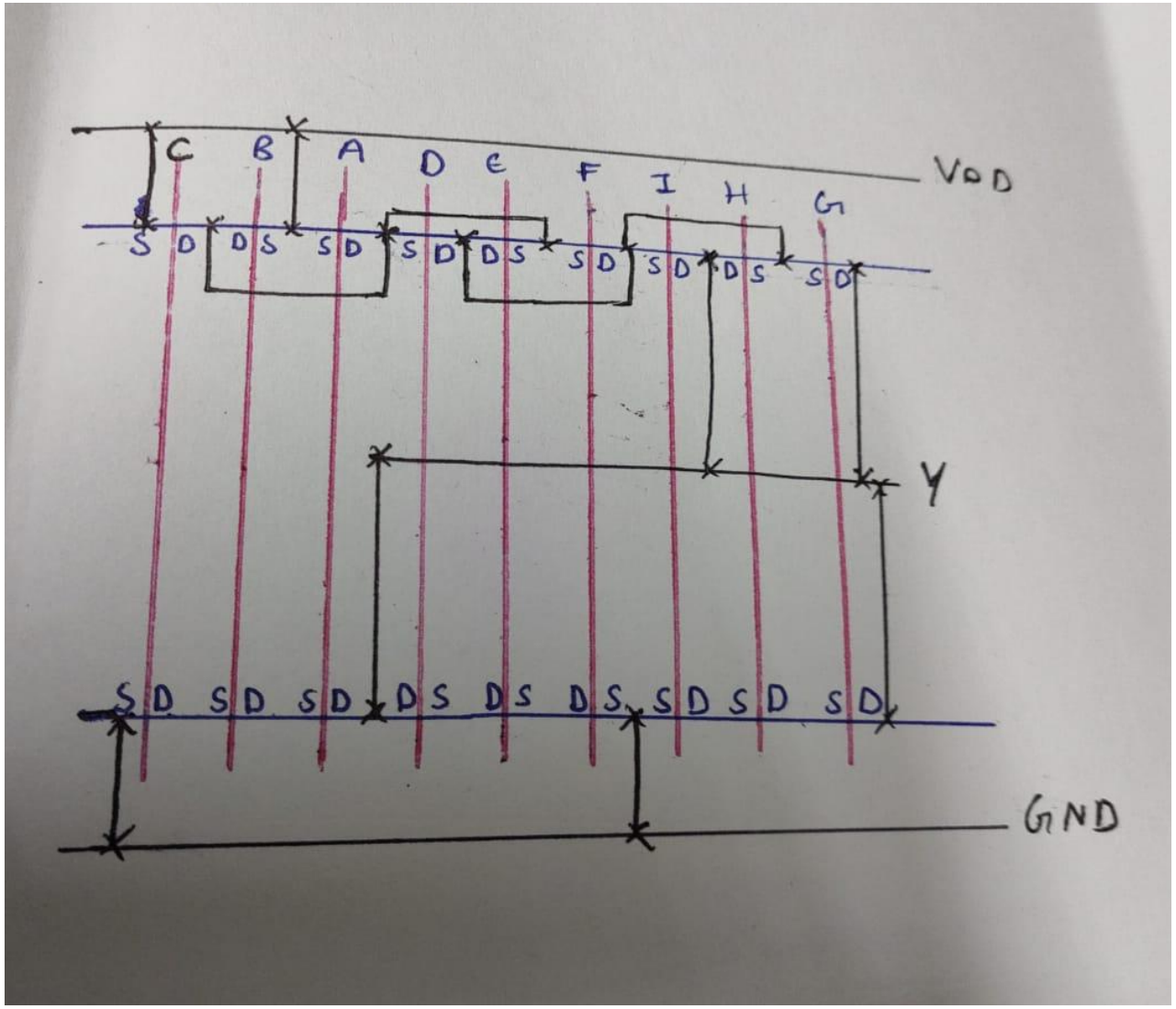
Schematic + Sizing (xCircuit)

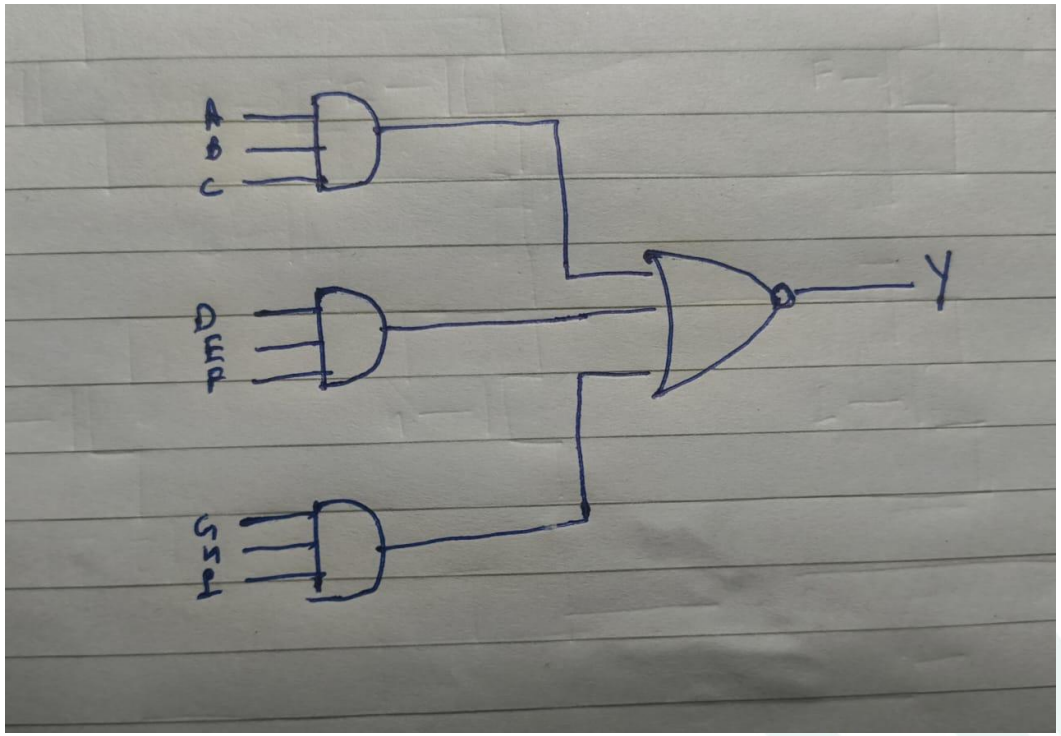


Sizing Information for Complex Gates

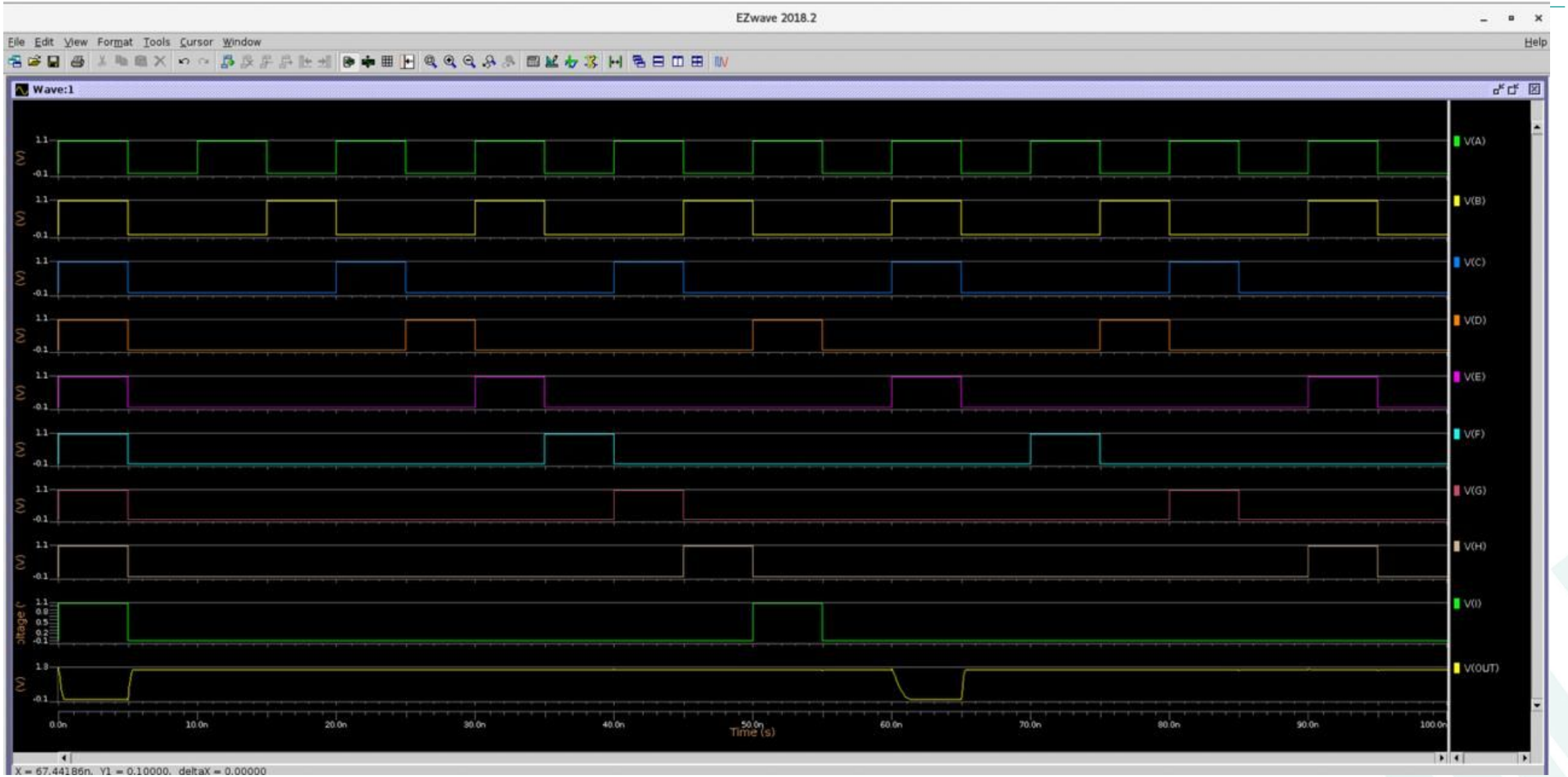
1) PMOS: W = 0.450μm, L = 0.06μm
2) NMOS: W = 0.225μm, L = 0.06μm

Stick Diagram and Logic Diagram

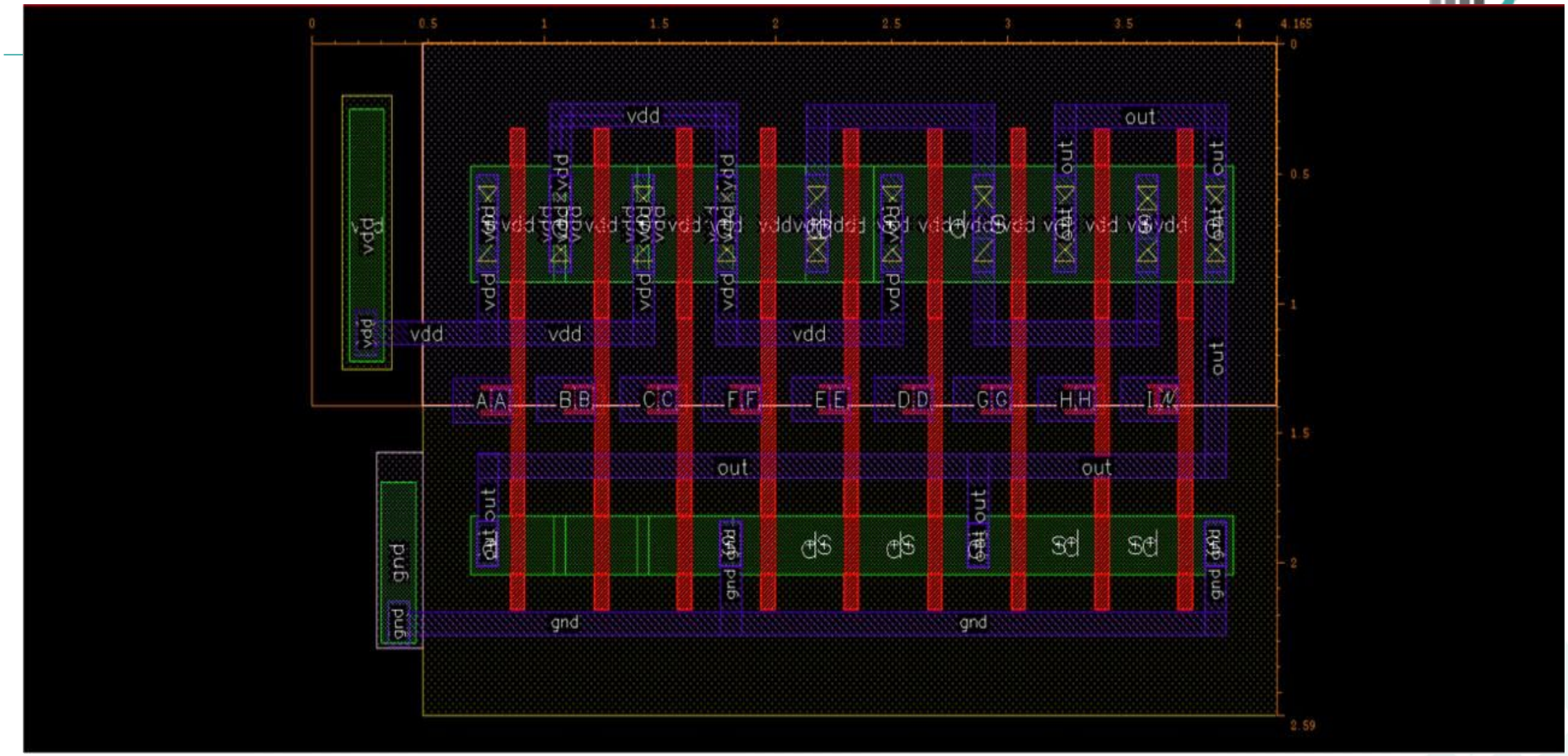




Pre Layout Complex

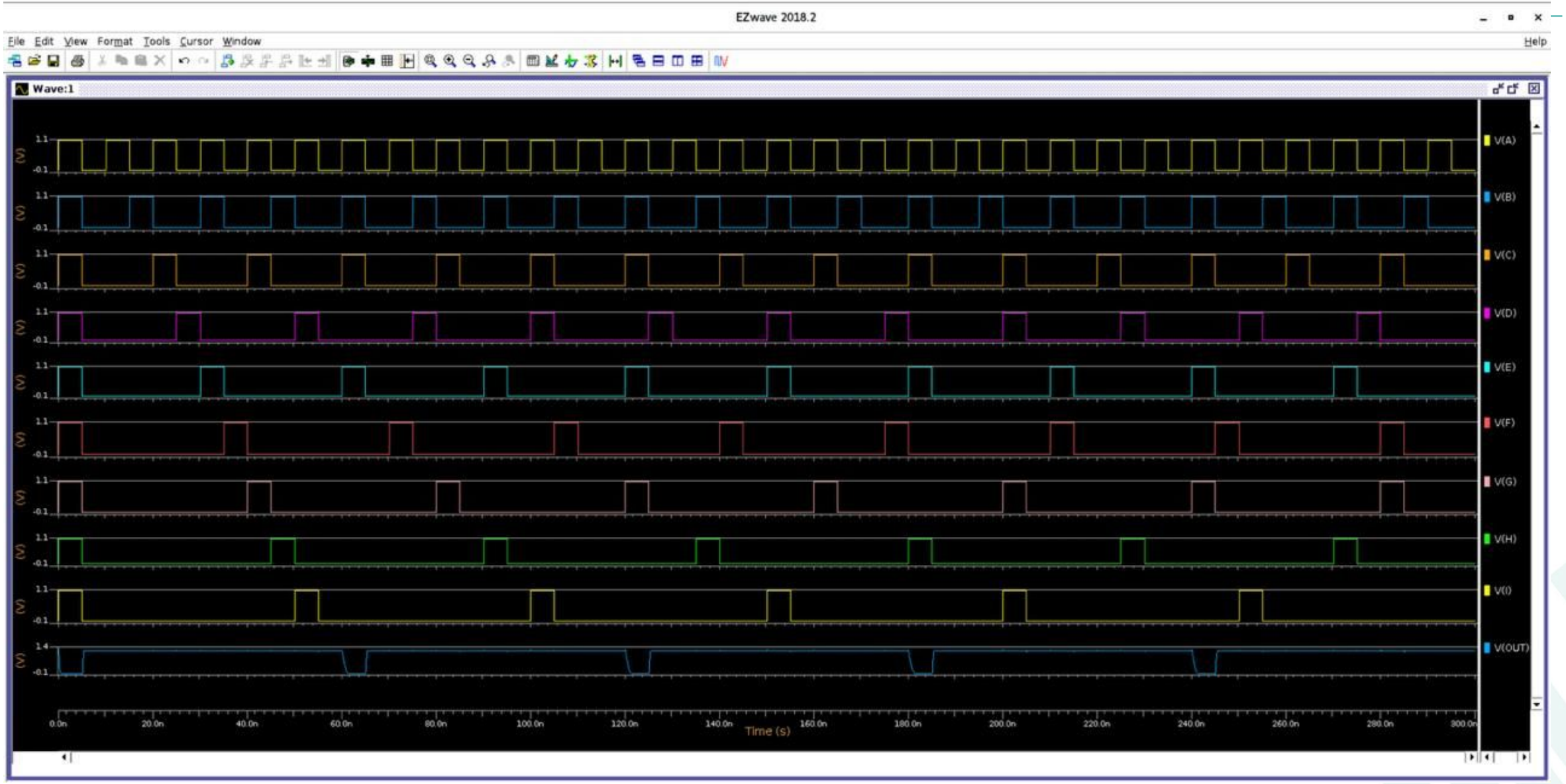


LAYOUT Complex

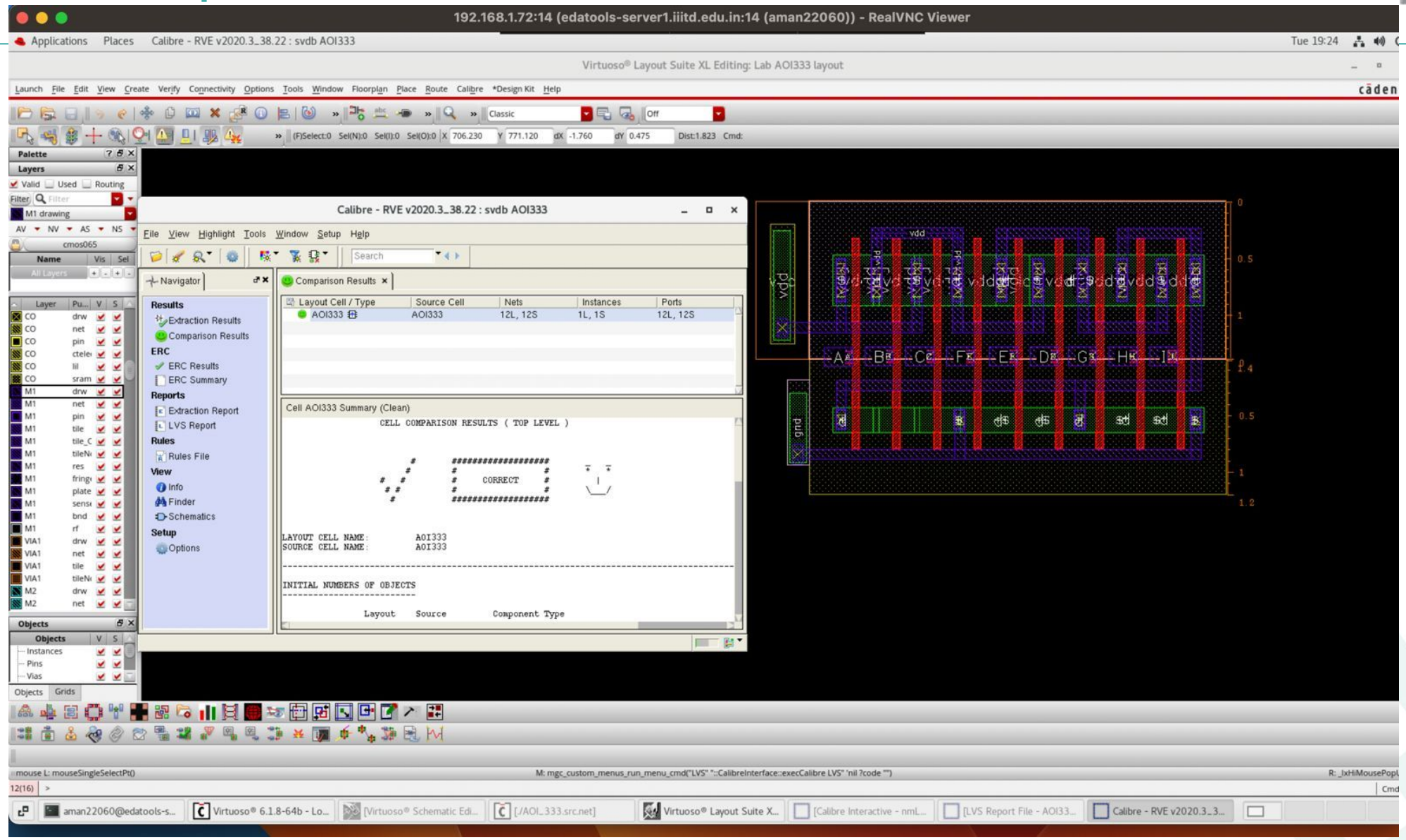


Length= 4.65 ,Mosfets=18 , Tracks=13
Area= 10.787 pico meter square

Post Layout Simulation Complex



LVS complex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Tue 19:24
Virtuoso® Layout Suite XL Editing: Lab AOI333 layout
Launch File Edit View Create Verify Connectivity Options Tools Window Floorplan Place Route Calibre *Design Kit Help
Calibre - RVE v2020.3_38.22 : svdb AOI333
File View Highlight Tools Window Setup Help
Navigator
Results
Extraction Results
Comparison Results
ERC
ERC Results
ERC Summary
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Comparison Results
Layout Cell / Type | Source Cell | Nets | Instances | Ports
AOI333 | AOI333 | 12L, 12S | 1L, 1S | 12L, 12S
Cell AOI333 Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
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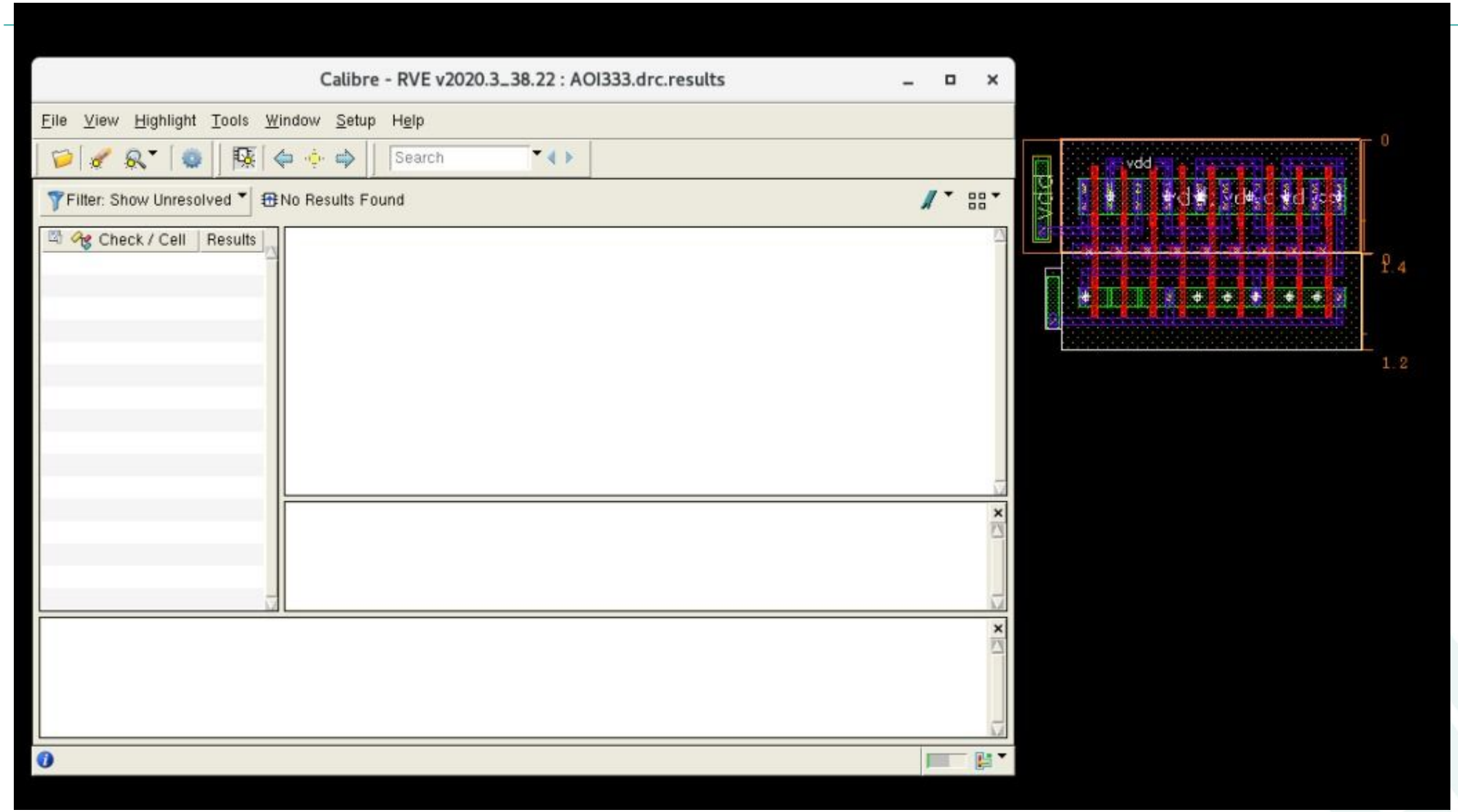
SOURCE CELL NAME: AOI333
INITIAL NUMBERS OF OBJECTS
Layout Source Component Type
vdd
gnd
A B C F E D G H I

DRC complex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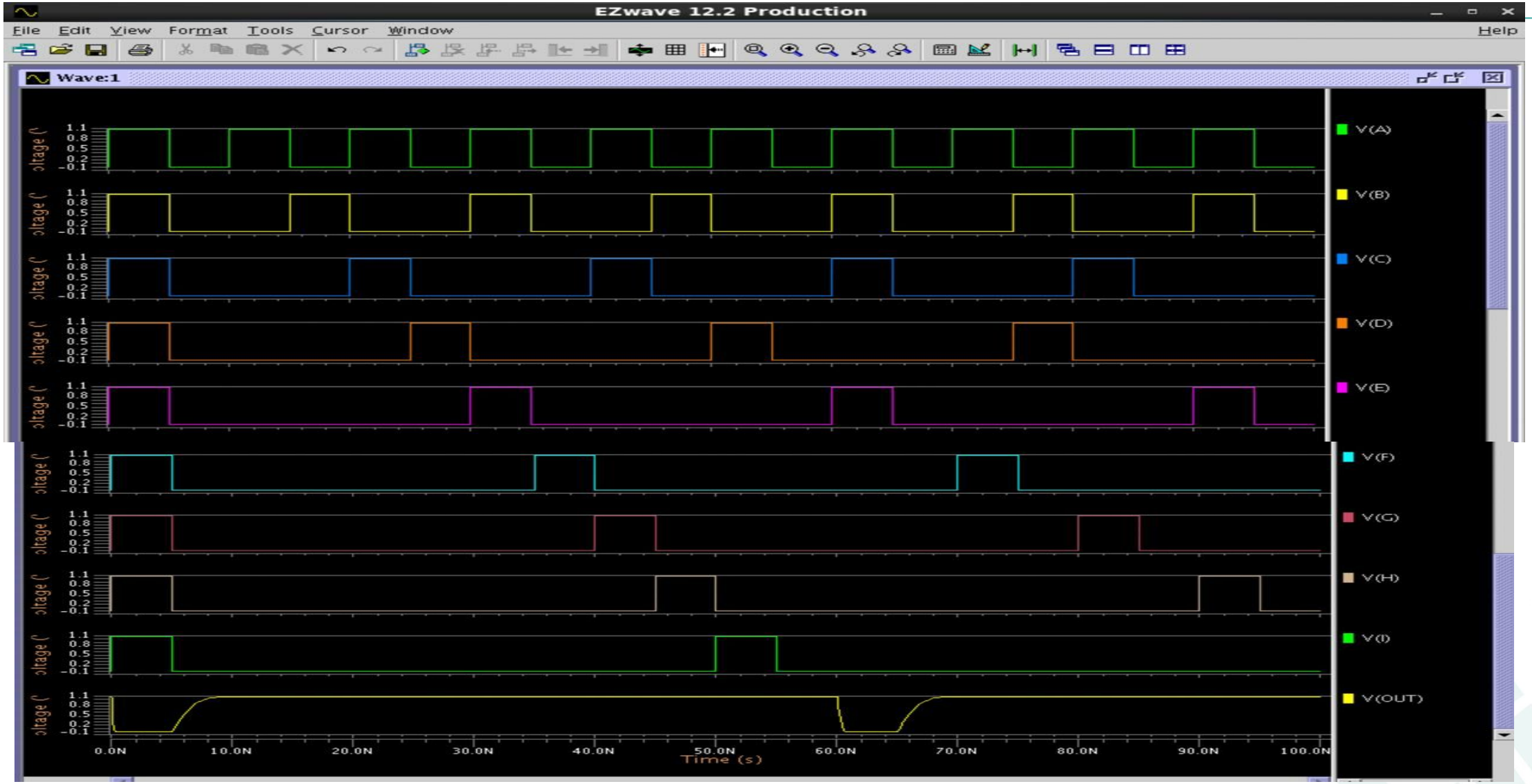
File View Highlight Tools Window Setup Help
Search
Filter: Show Unresolved
No Results Found
Check / Cell
Results
vdd
0
1.4
1.2

Pre Layout Non Complex



COMPLEX GATE VERIFICATION

PVT CONDITIONS	PARAMETER	PRE-LAYOUT	POST-LAYOUT
SS, 1.08,125	Fall Contamination Delay	2.491	2.407
	Fall Propagation Delay	1.863	2.086
	Rise Contamination	2.556	2.764
	Rise Propagation	1.875	2.064
SS, 1.08,25	Fall Contamination Delay	2.213	2.187
	Fall Propagation Delay	1.622	1.765
	Rise Contamination	2.508	2.617
	Rise Propagation	1.733	1.902

COMPLEX GATE VERIFICATION

PVT CONDITIONS	PARAMETER	PRE-LAYOUT	POST-LAYOUT
TT, 1.20,25	Fall Contamination Delay	1.868	1.988
	Fall Propagation Delay	1.375	1.455
	Rise Contamination	1.657	1.631
	Rise Propagation	1.187	1.561
FF, 1.32,125	Fall Contamination Delay	1.662	1.721
	Fall Propagation Delay	1.318	1.422
	Rise Contamination	1.392	1.421
	Rise Propagation	0.961	1.049
FF,1.32,-40	Fall Contamination Delay	1.179	1.357
	Fall Propagation Delay	1.005	1.062
	Rise Contamination	1.058	1.084
	Rise Propagation	0.749	0.840

NON-COMPLEX GATE VERIFICATION

PVT CONDITIONS	PARAMETER	PRE-LAYOUT	POST-LAYOUT
SS, 1.08,125	Fall Contamination Delay	0.802	
	Fall Propagation Delay	1.917	
	Rise Contamination	7.549	
	Rise Propagation	5.404	
SS, 1.08,25	Fall Contamination Delay	7.037	
	Fall Propagation Delay	1.730	
	Rise Contamination	6.854	
	Rise Propagation	4.97	

NON-COMPLEX GATE VERIFICATION

PVT CONDITIONS	PARAMETER	PRE-LAYOUT	POST-LAYOUT
TT, 1.20,25	Fall Contamination Delay	3.122	
	Fall Propagation Delay	1.954	
	Rise Contamination	3.885	
	Rise Propagation	2.112	
FF, 1.32,125	Fall Contamination Delay	3.462	
	Fall Propagation Delay	0.954	
	Rise Contamination	3.885	
	Rise Propagation	2.712	
FF,1.32,-40	Fall Contamination Delay	0.347	
	Fall Propagation Delay	0.776	
	Rise Contamination	3.027	
	Rise Propagation	2.120	

POWER CALCULATIONS(DYNAMIC)

COMPLEX

PVT CONDITIONS	PRE LAYOUT	POST LAYOUT
TT , 1.2 V, 25 C	0.760 mW	1.251 mW
FF, 1.32 v, -40 C	0.902 mW	2.259 mW
SS, 1.08 v, -40 C	0.611 mW	1.958 mW
SS, 1.08 v , 125 C	0.638 mW	1.718 mW

NON COMPLEX

PVT CONDITIONS	PRE LAYOUT	POST LAYOUT
TT , 1.2 V, 25 C	4.361 mW	
FF, 1.32 v, -40 C	4.421 mW	
SS, 1.08 v, -40 C	4.766 mW	
SS, 1.08 v , 125 C	4.308 mW	

POWER CALCULATIONS(STATIC)

COMPLEX

PVT CONDITIONS	PRE LAYOUT	POST LAYOUT
TT , 1.2 V, 25 C	4.79PW	6.32PW
FF, 1.32 v, -40 C	20.1PW	20.1PW
SS, 1.08 v, -40 C	61.1 PW	4.08PW
SS, 1.08 v , 125 C	273PW	200PW

NON COMPLEX

PVT CONDITIONS	PRE LAYOUT	POST LAYOUT
TT , 1.2 V, 25 C	13.8 PW	
FF, 1.32 v, -40 C	204 PW	
SS, 1.08 v, -40 C	476.6 PW	
SS, 1.08 v , 125 C	445PW	

Work Distribution

Aman Kumar : Schematic Designing(Virtuoso) of Complex Gate, eldo simulations,Post Layout of Complex , Pre Layout Simulation of Both ,Layout and Sizing

Sachin: Schematic (Xcircuit)

Rahul Kumar: Stick Diagrams,Schematic of Non Complex Gate